

Validity Analysis: INJONGO

Target context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Overall risk: **HIGH**

Deployment Context

Use case: Amharic utility chatbots for Ethiopian Electric Utility and Addis Ababa Water handling prepaid meter top-ups, outage reports, and bill disputes

Target population: Urban households in Addis Ababa and regional capitals interacting with utility support over Telegram and SMS

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	2	Significant gaps	high
Input Content 	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology 	2	Significant gaps	high
Output Content 	2	Significant gaps	high
Output Form 	4	Minor gaps	high
Average	2.7		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

INJONGO is a serious, ground-up effort to build a culturally grounded African intent detection and slot-filling benchmark, and Amharic is one of its 17 supported languages. Dataset inspection confirms that the Amharic split contains genuine Ethiopian-grounded content: Telebirr appears as PAYMENT_COMPANY, Ethiopian banks (Amara, CBE) appear as BANK_NAME, Ethiopian regional cities and calendar months populate slot values, and Ethiopic script renders cleanly. However, for the specific deployment of an Amharic-language utility chatbot for EEU and AAWSA on Telegram/SMS, three high-priority dimensions (input ontology, input content, output ontology, output content) show substantive validity gaps. The 40-intent taxonomy lacks the deployment's primary use cases (prepaid meter top-up, STS token entry, outage report,

load-shedding schedule). The 23-slot ontology lacks all deployment-critical slot types (METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, KEBELE). Code-switching — a confirmed high-priority deployment requirement — is absent from sampled Amharic data. Annotation was performed by pan-African crowdworkers without documented utility domain expertise, against a deployment requirement that ground truth be validated by EEU/AAWSA agents. Strongest dimensions are input form (text Ethiopic Unicode matches Telegram exactly) and output form (text intent+slots, standard metrics). The benchmark is best used as a fine-tuning starting point for general Amharic intent detection rather than a deployment-readiness evaluation for utility-specific tasks.

Practical Guidance

What This Benchmark Measures

INJONGO measures general-purpose Amharic intent detection (across banking, home, travel, kitchen & dining, and a generic utility domain) and slot-filling for 23 broad named-entity types. For the Ethiopian utility deployment, it can evaluate baseline Amharic NLU competence — whether a model recognizes Ethiopian banks, Telebirr, Ethiopian cities/calendar, formal polite Amharic register, and electricity/water bill_balance queries. Strongest dimensions are input form (Ethiopic Unicode in text matches Telegram deployment exactly) and output form (standard intent accuracy + slot F1 metrics).

Construct Depth

Construct depth is shallow for the deployment task. The benchmark probes general conversational NLU with cultural grounding but does not probe the specific constructs the deployment requires: prepaid meter workflows, STS token recognition, outage reporting, load-shedding queries, kebele-level address resolution, or Amharic-English code-switched technical vocabulary. The benchmark's 80-per-intent uniform distribution does not reflect real EEU/AAWSA traffic skew, so even on the intents that exist, the evaluation does not measure performance under deployment-realistic frequencies. Annotation is by general native speakers, not utility domain experts, so slot-boundary correctness for utility billing terminology is not validated against the authoritative ground-truth standard.

What Else You Need

Substantial supplementation is required. (1) Augment intent taxonomy with deployment-specific intents (input_ontology gap) using EEU/AAWSA conversation logs or expert-elicited examples. (2) Augment slot ontology with METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, KEBELE, WOREDA, sub-city types (output_ontology gap). (3) Collect or synthesize Amharic-English code-switched utterances for utility domain technical vocabulary (input_content gap), since INJONGO's monolingual Amharic data will overestimate deployment performance. (4) Re-annotate or validate utility-specific slot boundaries with EEU/AAWSA customer service agents (output_content gap). (5) Build a deployment-distribution test set reflecting real intent frequencies, not uniform 80-per-intent. (6) Verify Unicode normalization (NFC/NFD) compatibility between INJONGO release files and the production tokenizer pipeline.